

Square Values and Square Roots

Teacher's Edition

Read This First

This lesson differs from the Math Nation version on a point of mathematics, not of style.

Math Nation's Lesson 2 teaches that $\sqrt{16} = \pm 4$. It is on their guided instruction page, on both of their practice sheets, and in the answer key to every square-root question in the lesson. It is wrong. The radical symbol denotes the **principal** — the non-negative — square root. $\sqrt{16} = 4$. The equation $x^2 = 16$ is what has two solutions, $x = \pm 4$.

Their own materials show the strain. The teacher note on their guided-practice page says students “will be introduced to the principal square root in a future lesson,” so the definition on these pages is one they intend to retract. And their Xavier item points a student wrong for answering $\sqrt{25} = 5$, which is correct.

Three other benchmarks in this course require the symbol to have exactly one value.

MA.8.NSO.1.7 places radicals inside order-of-operations problems — if $\sqrt{16}$ were ± 4 , then $3 + \sqrt{16}$ would be 7 or -1 and the item would be unanswerable. MA.8.NSO.1.2 asks students to plot square roots on a number line, which a two-valued object cannot be. MA.8.GR.1.1 uses a square root for the hypotenuse, and a length of ± 5 is not a length.

MA.8.AR.2.3's own clarification reads “focus on understanding that solving $x^2 = p$ gives both a positive and a negative solution.” That is a statement about the equation. Florida never attaches the $\pm$ to the symbol.

Teaching it correctly costs one extra line and buys this lesson a real difficulty in place of a false rule. Notes II (slide 4) and Example 2 (slides 10–12) are where it lives.

Standard and Targets

MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.

Clarification Focus on understanding that solving $x^2 = p$ gives both a positive and a negative solution.

Boundary Square roots of perfect squares up to 225. Students are not required to isolate the x^2 term.

Learning targets I can name the parts of a radical. I can evaluate a square root. I can find every real solution of $x^2 = p$, including saying when there is none.

Essential questions How many solutions does $x^2 = p$ have? What does the radical symbol ask for? Why are those two different questions?

Building on Lesson 1 — squaring, perfect squares, and finding a side from an area.

Working toward MA.8.NSO.1.1 in Lesson 4, where square roots of non-perfect squares get estimated and named irrational, and MA.8.GR.1.1 in Unit 6.

Where this sits in the course Second of three consecutive lessons on MA.8.AR.2.3, which the Grade 8 FAST reports under Algebraic Reasoning. Math Nation also lists MA.8.NSO.1.1 on this

lesson, but nothing here approximates a square root of a non-perfect square — that benchmark genuinely arrives in Lesson 4, and the tag is aspirational.

Materials Whiteboards and markers. Calculators are allowed but every radicand in this lesson is on Lesson 1's list or is 0 or negative, and none of those needs one. Worksheet 7A.02.02. The perfect-square anchor chart from Lesson 1, still on the wall.

Lesson at a Glance

Windy Hill periods run **53 minutes**.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up: same and different	2	4 min
Notes I–IV: the parts of a radical, two questions, the sign outside, three cases	3–6	14 min
Example 1: $\sqrt{144}$ and $x^2 = 144$	7–9	5 min
Example 2: Xavier is right twice	10–12	5 min
Whiteboards: 9 questions, each with its reveal	13–30	19 min
IXL — start today's skills	31	5 min
Total		53 min

Where the lesson lives: Notes II (slide 4) and Example 2 (slides 10–12). Notes II is the distinction, stated. Example 2 is the distinction, felt — students meet a paper where both answers are right and have to work out why they differ.

Read this before you plan the day. Math Nation's component times total 45 to 50 minutes before any worksheet. The plan above fits 53 because their Guided Instruction and Guided Practice are compressed into Notes II and III with Examples 1 and 2 as the worked versions, and the Try It and Your Turn! become the whiteboard rounds.

No partner or group work anywhere in this lesson. Math Nation's version has four printed instances plus the Compare and Connect routine. All four are “discuss with your partner” prompts attached to questions students can answer alone; they are now teacher-led questions on the corresponding slide.

Vocabulary

radical — the symbol $\sqrt{\quad}$ that asks for a root

radicand — the number underneath the radical

index — the small number in the notch, saying which root to take

square root — a number that gives the radicand when multiplied by itself

principal square root — the positive one — the value the radical symbol returns

A note on the glossary. Math Nation lists five additional terms and no key terms here. Four of theirs are above. The fifth is “perfect square,” which was Lesson 1’s. “Principal square root” has been added, because it is the name of the thing this lesson is about and their materials postpone it to Lesson 4 while teaching the opposite in the meantime.

Slide-by-Slide Teaching Notes

Slide 1. Square Values and Square Roots *1 min*

- Post the learning target.
- Frame it exactly this way: yesterday every answer was a length, so every answer was positive.
- Today the question changes from 'find the side' to 'solve the equation', and that change is what brings in the negative answer.
- The one thing students must leave with is that the radical symbol and the equation ask different questions.
- Say that sentence today and say it again tomorrow.

Slide 2. Warm-Up *4 min*

- TYPE: Same and different.
- INDEPENDENT.
- Expect: same — both involve a number squared, and 36 and 25 are both perfect squares from yesterday’s list.
- Different — one is a symbol to evaluate and one is an equation to solve, and (this is the one to fish for) one has a single answer and the other has two.
- Do not supply that last observation; if nobody offers it, leave it and Notes II will land harder.
- This replaces Math Nation’s deep-sea clam estimation problem, which was sound arithmetic with no connection to square roots.
- OFF THE SLIDE, YOURS TO SAY: Both are about squaring. One asks for a value; the other asks which values work.

Slide 3. Notes: I. The parts of a radical *3 min*

- Students copy the three words and the labeled diagram.
- Say the words out loud several times — Florida lists 'radical' in the grade 8 terminology and the FAST will use it.
- The one thing to insist on: the missing index is not missing, it is a 2 that nobody writes, the same way nobody writes the 1 in $1x$.
- An index of 3 arrives tomorrow and every index other than 2 gets written down.
- OFF THE SLIDE, YOURS TO SAY: When no index is written, the index is 2 and it means a square root.

Slide 4. Notes: II. The symbol and the equation ask different questions *4 min*

- THE NEW IDEA IN THIS LESSON, and the place where these materials differ from the book on purpose.
- Math Nation’s pages teach that the square root of 25 is plus-or-minus 5.

- That is not how the symbol works anywhere else in this course: MA.8.NSO.1.7 puts radicals inside order-of-operations problems, MA.8.NSO.1.2 asks students to plot square roots on a number line, and MA.8.GR.1.1 uses one for the hypotenuse — none of those work if the symbol has two values at once.
- Build the table on the board, left column then right, and say the framing sentence: the equation asks how many numbers work, and there are two; the symbol asks for one number, and it hands you the positive one.
- OFF THE SLIDE, YOURS TO SAY: $\sqrt{25} = \pm 5$ is the error this lesson exists to prevent.

Slide 5. Notes: III. The sign lives outside the symbol *3 min*

- Students copy all three lines of item B, spaced out, because the shape on the page is the lesson.
- Ask which of the three is a single number (all of them, except the last, which is a pair).
- The line students call wrong out of caution is $-\sqrt{25}$ — it is correct, it is standard, and it is the notation they will need for the quadratic formula in three years.
- OFF THE SLIDE, YOURS TO SAY: $-\sqrt{25}$ is not an error. It is how you write the negative root when you want it.

Slide 6. Notes: IV. Solving $x^2 = p$ — three cases *4 min*

- Students copy the table.
- The ZERO row is the one they forget: they answer ± 0 , which is not so much wrong as meaningless — 0 and -0 are the same number, so there is one solution and not two.
- The NEGATIVE row needs the reason in the panel, not just the rule.
- Write 'no real solution' as the required phrase; do not accept a blank, do not accept 0, do not accept -7 .
- If a student asks what the non-real answers are, tell them those are imaginary numbers, they exist, and they are a high-school topic.
- OFF THE SLIDE, YOURS TO SAY: Nothing real squares to a negative. Two positives make a positive — and so do two negatives.

Slide 7. Example 1 *1 of the 5 min*

- SPOKEN ONLY.
- No pencils.
- The right-hand one has two.
- Take a wrong answer before you take a right one — a student who says the left has two is reading the symbol the way Math Nation's page does, and you want that said aloud today rather than discovered on the test.

Slide 8. Example 1 *2 of the 5 min*

- Work the radical first and the equation second, in that order.
- Then read both out loud in plain English: 'what is the square root of 144' has one answer, and 'what values of x make x squared equal 144' has two. 144 is on yesterday's perfect-square list, so nobody should be reaching for a calculator.
- OFF THE SLIDE, YOURS TO SAY: One line asks for a value. The other asks which values work. The question tells you which.

Slide 9. Example 1 *2 of the 5 min*

- Substitute both, every time, this week.
- It costs fifteen seconds and it is the only thing that makes students believe –12 genuinely works rather than being a rule they were told.
- The panel is the sentence to repeat: checking proves the equation has two solutions, and it says nothing at all about what the symbol means.
- OFF THE SLIDE, YOURS TO SAY: Both really do work. That is why the equation has two answers and the symbol still has one.

Slide 10. Example 2 *2 of the 5 min*

- Ask the question as printed and take a vote.
- Most classes pick the right-hand one, because 25 appears twice and the answers differ.
- Do not resolve it.
- The next slide resolves it, and the vote is what makes them watch.

Slide 11. Example 2 *1 of the 5 min*

- Say it plainly and let it sit.
- Math Nation prints this item with Xavier marked as getting one of the two wrong — and the one they mark wrong is the one that is actually correct.
- Do not repeat that.
- The productive confusion is real: students feel that the same number 25 ought to give the same answer both times, and naming that feeling before resolving it is what makes the resolution stick.

Slide 12. Example 2 *2 of the 5 min*

- Let the class supply these lines rather than reading them out.
- Worksheet question 12 is this item with Nadia's name on it and only one statement, and question 11 is the select-all version, so the wording you accept here is the wording you will read back tonight.
- OFF THE SLIDE, YOURS TO SAY: If somebody now says “so $\sqrt{25}$ is both 5 and -5 ”, go back to Notes II and read the sentence again.

Slide 13. Whiteboards • Question 1 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 14. Whiteboards • Question 1 of 9 — reveal *about 1 minute with the discussion*

- The baseline: the radical, on its own. 8, not ± 8 .
- Everything after this changes exactly one thing, so a wrong answer later names the thing that changed.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 15. Whiteboards • Question 2 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 16. Whiteboards • Question 2 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: an equation instead of the symbol.
- Same 64 underneath, and now there are two answers.
- A board saying 8 has answered question 1 again — that student is reading the equation as if it were the symbol, which is Example 2 exactly.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 17. Whiteboards • Question 3 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 18. Whiteboards • Question 3 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: a minus sign in front. $-\sqrt{64} = -8$, one number, and it is negative because you put it there.
- Expect boards saying 'no real value' — that is the minus being read as if it were under the radical, which is question 6's job, not this one's.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 19. Whiteboards • Question 4 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 20. Whiteboards • Question 4 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the right-hand side is now 0.
- ONE solution, not two.
- Boards saying ± 0 are the ones to talk about — 0 and -0 are the same number, so writing $\pm$ in front of it says nothing.
- Notes IV, middle row.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 21. Whiteboards • Question 5 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 22. Whiteboards · Question 5 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the sign of p.
- Now there are none.
- Both 8 and -8 square to positive 64, so there is no third option left to try.
- A blank board, a 0 and a -8 are all wrong; the answer is words.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 23. Whiteboards · Question 6 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 24. Whiteboards · Question 6 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT CLOSSES THE LOOP.
- One feature changed from question 5: back to the symbol, same negative underneath.
- Same refusal, and students who got question 5 right often hesitate here because the shape on the page changed.
- Ask what is different between questions 5 and 6 — nothing that matters, and that is the answer.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 25. Whiteboards · Question 7 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 26. Whiteboards · Question 7 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: a new number, and the question is back to the equation form.
- Two answers, the same as question 2.
- This is the retired worksheet's question 2, where the two boxes sit side by side for exactly this reason — the radical on its own and the equation are different questions about the same 25.
- A board saying 5 is answering the radical question; ask it what it did with the minus five.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 27. Whiteboards · Question 8 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 28. Whiteboards · Question 8 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT EARNS ITS PLACE, and it is worth the whole round.
- One feature changed: a fraction.

- YESTERDAY THIS EXACT LINE HAD ONE ANSWER — lesson 2.01's last whiteboard question showed $s^2 = 4/9$ and the answer was $2/3$, because s was the side of a square and a side is a length.
- Today nobody said it was a length, so there are two.
- Same symbols, same arithmetic, different answer, and the only thing that changed is what the letter is allowed to be.
- Put the two slides up together if you still have yesterday's deck.
- A board saying $2/3$ is not doing worse mathematics than one saying $\pm 2/3$ — it is answering yesterday's question, and that is a far more interesting thing to say out loud than 'wrong'.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 29. Whiteboards · Question 9 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 30. Whiteboards · Question 9 of 9 — reveal *about 2 minutes with the closing discussion*

- WRITTEN WORK, and the only question in the round that asks for it.
- With the worksheet gone this is the only place in the period the habit gets built, and IXL takes an answer without ever seeing the work.
- ANSWER: no, $\sqrt{64} = 8$.
- THE MARK IS FOR SEPARATING THE TWO CLAIMS, because Jordan's reason is TRUE and his conclusion is false: 8 and -8 really do both square to 64, and the radical symbol still asks for one number — the positive one.
- A student who writes only "no, it is 8" has the right answer and has not engaged with the only interesting thing in the question.
- The equation $x^2 = 64$ is where -8 belongs, and it has been on this board twice today.
- Say it as a rule they can carry: the SYMBOL answers with one number, the EQUATION answers with every number that works.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name.

Slide 31. IXL *5 min*

- IXL.
- The skills are assigned in Google Classroom, not named here, and they are deliberately not ours -- Croix: "The students need questions that aren't exclusively mine."
- Makes them think." Circulate.
- The thing to watch for is a student answering in the box with no paper in front of them: IXL takes an answer and never sees the work, so this is the one part of the period where the habit can quietly lapse.

Worksheet 7A.02.02 — Answer Key

1. the diagram	<i>index (top left) · radical (the symbol) · radicand (under the symbol)</i> No index written means an index of 2, which means a square root.
2. the two boxes	<i>Left: $\sqrt{25} = 5$ Right: $x = \pm\sqrt{25} = \pm 5$</i> A ± 5 in the left box is the error this lesson exists to prevent. Both boxes correct is the target.
3. $\sqrt{49}$	<i>7</i>
4. $\sqrt{121}$	<i>11</i>
5. $\sqrt{0}$	<i>0</i> Zero is the one radicand with a single square root rather than a positive and a negative one.
6. $\sqrt{-36}$	<i>No real value.</i> Not 0 and not -6. Nothing real squares to a negative.
7. $x^2 = 64$	<i>$x = \pm 8$</i> $8 \times 8 = 64$ and $(-8) \times (-8) = 64$.
8. $144 = m^2$	<i>$m = \pm 12$</i> The variable on the right changes nothing.
9. $x^2 = 0$	<i>$x = 0$</i> One solution, not two.
10. $y^2 = -49$	<i>No real solution.</i> Same refusal as question 6, asked about an equation. Students should be able to say which is which.
11. select all	<i>9, -9, $\sqrt{81}$ and $-\sqrt{81}$</i> Four of six, and two of the four are the same numbers written differently. 40.5 is $81 \div 2$, the halving error; 81^2 is 6,561.
12. Nadia	<i>$\sqrt{36} = 6$, and $x^2 = 36$ has two solutions, $x = \pm 6$.</i> Nadia is not confused about arithmetic — she knows both 6 and -6 square to 36. She has attached the $\pm$ to the wrong object. Say that out loud when you go over it. Both statements are needed for the point.

Question 11 is the one to circulate for. Four of the six boxes are correct, and two of the four — $\sqrt{81}$ and $-\sqrt{81}$ — are 9 and -9 written a different way. A student who ticks the two plain numbers and leaves the radicals blank has the arithmetic and not the notation, which is exactly what the FAST's select-all items are built to catch.

Question 12 is the one to read tonight. It tells you who owns slide 4. Accept any wording that separates the symbol from the equation; do not accept an answer that only fixes the arithmetic.

Additional Practice 7A.02.02 — Answer Key

1. the diagram	<i>index · radical · radicand</i> An index of 2 is never written. Any other index always is.
2. the two boxes	<i>Left: $\sqrt{16} = 4$ Right: $x = \pm\sqrt{16} = \pm 4$</i>
3. $\sqrt{64}$	<i>8</i>
4. $\sqrt{225}$	<i>15</i> 225 is the largest perfect square the Grade 8 FAST will ask about.
5. $\sqrt{100}$	<i>10</i>
6. $\sqrt{-16}$	<i>No real value.</i> Not 0 and not -4.
7. $x^2 = 25$	<i>$x = \pm 5$</i>
8. $196 = a^2$	<i>$a = \pm 14$</i>
9. $n^2 = 0$	<i>$n = 0$</i> One solution, not two.
10. $x^2 = -100$	<i>No real solution.</i>
11. select all	<i>12, -12, $\sqrt{144}$ and $-\sqrt{144}$</i> Four boxes, two numbers. 72 is $144 \div 2$ and 288 is 144×2 .
12. Owen	<i>$\sqrt{100} = 10$, and $x^2 = 100$ has two solutions, $x = \pm 10$.</i> Arithmetic right, notation wrong. Both halves are needed for the point.

This sheet is optional. Post it to Google Classroom for students who want more. It is not assigned and it is not collected.

Changes from the Math Nation Lesson

- **The definition of the radical symbol was corrected.** Math Nation teaches $\sqrt{16} = \pm 4$ on their guided instruction page, in both practice sets and throughout the answer key. The radical denotes the principal — non-negative — square root, and the $\pm$ belongs to the equation $x^2 = p$. Every square-root answer in these materials is the principal root, and slide 4 makes the distinction the lesson. See “Read This First” on page 1.
- **The Xavier item was rebuilt.** As printed, Xavier answers $b^2 = 25$ with 5 and -5 and answers $\sqrt{25}$ with 5, and the key points him as having got one of the two right. Under the correct definition he is right both times. The item is now slides 10–12, where both answers are affirmed and the class explains why they differ — which is a better question than the original was.
- **A false teacher note was dropped.** Math Nation’s guided instruction tells the teacher to “explain that the index can be any rational number.” The index of a radical is a whole number of at least 2. The note now says so.
- **An inconsistent generalization was repaired.** Their boxed rule reads “for equations in the form $x^2 = p$, where p is a whole number,” and the very next question uses $x^2 = -36$. Whole numbers are not negative. Notes IV now states the three cases by the sign of p , which covers the negative case without contradicting itself.
- Every partner and group activity is replaced — four printed instances plus the Compare and Connect routine. All were “discuss with your partner” prompts on questions a student can answer alone; they are now teacher-led questions on the slide.
- The warm-up was replaced. Math Nation’s deep-sea clam estimation problem is arithmetically sound but has no connection to square roots. The new warm-up is a same-and-different on $\sqrt{36}$ and $x^2 = 25$: both are about squaring, but one names a value and the other asks which values work. That distinction is the whole lesson, and it is on the board before a word is said.
- The wrap-up was replaced. Math Nation’s asks students to complete two of five sentence starters, all keyed “answers will vary.” The Ticket Out the Door asks for a value, a solution set, and one sentence explaining why they differ.
- A select-all item was added, in the format the FAST uses and the Math Nation materials never do.
- Answer keys were built for both sheets from the same source file as the student versions, so the two cannot drift apart.
- **Nothing else in the mathematics was changed.** Every value, every radicand and every equation in the original practice sets is still here; what changed is what the symbol is said to mean.